

Calculus I Lesson 16

1) $f(x) = 6$

Hint: $f(x + \Delta x) = 6$ and $\lim_{x \rightarrow 0} k = k$ where k is constant.

a) $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

2) $f(x) = -3$

Hint: $f(x + \Delta x) = -3$ and $\lim_{x \rightarrow 0} k = k$ where k is constant.

a) $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

3) $f(x) = c$ where c is constant.

Hint: $f(x + \Delta x) = c$ and $\lim_{x \rightarrow 0} k = k$ where k is constant.

a) $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

4) $f(x) = x$

a) $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \frac{x + \Delta x - x}{\Delta x} = \underline{\quad ? \quad}$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\quad ? \quad}$

5) $f(x) = x^2$

Hint: $(a + b)^2 = a^2 + 2a \cdot b + b^2$

a) $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \frac{(x + \Delta x)^2 - x^2}{\Delta x} = \frac{x^2 + 2x \cdot \Delta x + (\Delta x)^2 - x^2}{\Delta x} = \underline{\hspace{2cm}}?$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}}?$

6) $f(x) = x^3$

Hint: $(a + b)^3 = a^3 + 3a^2 \cdot b + 3a \cdot b^2 + b^3$

a) $\frac{f(x + \Delta x) - f(x)}{\Delta x} = \frac{(x + \Delta x)^3 - x^3}{\Delta x} = \frac{x^3 + 3x^2 \cdot \Delta x + 3x \cdot (\Delta x)^2 + (\Delta x)^3 - x^3}{\Delta x} = \underline{\hspace{2cm}} ?$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

$$7) f(x) = x^4$$

$$\text{Hint: } (a + b)^4 = a^4 + 4a^3 \cdot b + 6a^2 \cdot b^2 + 4a \cdot b^3 + b^4$$

$$\text{a) } \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \frac{(x + \Delta x)^4 - x^4}{\Delta x}$$

$$= \frac{x^4 + 4x^3 \cdot \Delta x + 6x^2 \cdot (\Delta x)^2 + 4x \cdot (\Delta x)^3 + (\Delta x)^4 - x^4}{\Delta x} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$\text{b) } f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

8) $f(x) = x^5$

Hint: $(a + b)^5 = a^5 + 5a^4 \cdot b + 10a^3 \cdot b^2 + 10a^2 \cdot b^3 + 5a \cdot b^4 + b^5$

a) $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

$$= \frac{(x + \Delta x)^5 - x^5}{\Delta x}$$

$$= \frac{x^5 + 5x^4 \cdot \Delta x + 10x^3 \cdot (\Delta x)^2 + 10x^2 \cdot (\Delta x)^3 + 5x \cdot (\Delta x)^4 + (\Delta x)^5 - x^5}{\Delta x} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm} ? \hspace{2cm}}$

$$9) f(x) = x^n$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{1cm}} ?$$

Power Rule for derivative of a function

